

ALLEN DEVARAJ AUGUSTIN PONRAJ

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EDUCATION

University of Colorado Boulder

Boulder, CO, USA

M.S. in Aerospace Engineering Sciences (GPA: 3.9/4.0)

Aug 2025 – May 2027

- Coursework: Linear Control Systems, Algorithmic Motion Planning, Statistical Estimation

Madras Institute of Technology, Anna University

Chennai, Tamil Nadu, India

B.E. in Aeronautical Engineering (GPA: 3.51/4.0)

Nov 2021 – May 2025

- Coursework: Rocket Propulsion, Aircraft Structures, Avionics Systems

TECHNICAL SKILLS

CAD Design: Autodesk Fusion 360, CATIA, SolidWorks

PCB Design: Autodesk Eagle, Proteus

Programming Languages: Python, C++

Microcontrollers & Embedded Systems: Arduino IDE, Raspberry Pi, Pixhawk

Robotics Frameworks & Environment: ROS2, Gazebo, MuJoCo, MuJoCoMPC

Simulation: Autodesk CFD

Data Analysis: MATLAB, Mission Planner, QGroundControl

Manufacturing Processes: Soldering, 3D Printing

EXPERIENCE

Graduate Research Assistant

Boulder, CO, USA

Correll Lab – Intelligent Robotics Laboratory

Jan 2026 – Present

Humanoid Whole-Body Control (Unitree H1-2)

- * Developed a sampling-based model predictive control framework using the Cross-Entropy Method (CEM) for online receding-horizon whole-body planning, replanning every control cycle to maintain balanced standing without offline demonstrations or precomputed reference trajectories.
- * Extended the planner to contact-implicit MPC that discovers bracing contacts at runtime, enabling the humanoid to lean on a table with one arm to extend its stable reach envelope while manipulating with the other, with no pre-specified contact schedule.

Robotic Disassembly for Battery Recycling (ARPA-E, DoE)

- * Developed a constrained motion planning pipeline with analytical IK seed ranking and cost-normalized configuration selection for a 7-DOF industrial manipulator system.

ADCS Engineer

Boulder, CO, USA

Colorado Center for Astrodynamics Research

Sept 2025 – Present

- Working on the Attitude Determination and Control Systems (ADCS) subsystem of the CubeSat mission SWARM-EX funded by the NSF.
- Developing an interpreter between the Blue Canyon Technologies ADCS component XACT-15 and the CDH subsystem using Hydra-based communication architecture for ground station integration.

Project Intern – ADCS Testbed for CubeSats

Trivandrum, Kerala, India

SSPACE Lab, Indian Institute of Space Science and Technology

Jun 2024 – Jul 2024

- Developed and implemented a one degree-of-freedom control algorithm for a single reaction wheel on a FloatSat with air bearings to test CubeSat ADCS systems.
- Achieved precise wireless Z-axis stabilization and control for the FloatSat platform.

PROJECTS

SLAM with DeepRacers

Feb 2026 – May 2026

- Implemented SLAM for an autonomous rover navigating underground corridors using LiDAR, IMU, and odometry without wheel encoders.
- Developed a global A* path planner integrated with a Pure Pursuit geometric controller for navigation.
- Built a vision-based line-following pipeline using an Intel RealSense camera and image segmentation techniques.

Belief-Augmented RL for Pick-and-Place under Uncertainty

Jan 2026 – Apr 2026

- Evaluated standard PPO and belief-based PPO reinforcement learning approaches under varying uncertainty and noise conditions.
- Simulated uncertainties including distance-dependent sensor noise, occlusion-induced observation dropout, and workspace drift noise.
- Developed a robotic manipulation environment where an SO101 robotic arm autonomously picks a red LEGO block from mixed blue blocks without a vision pipeline.

Dual Manipulators Performing Insertion Task using TAMP

Aug 2025 – Dec 2025

- Performed a coordinated peg-in-hole insertion task using two Franka manipulators through Task and Motion Planning (TAMP) in a SITL environment.

Methods of Stabilization in Ground Effect Aircrafts

Aug 2024 – May 2025

- Engineered stabilization methods for Wing-In-Ground Effect Aircrafts via CFD-based SOGA optimization of tail geometry, achieving a 9.468% increase in lift curve slope.
- Designed a feedback control system for pitch damping to improve longitudinal aircraft stability.
- Contributed to the design and fabrication of a remote-controlled aircraft for testing.

AeroTHON 2024 – Team Classified

Mar 2024 – Nov 2024

- Led a team in an SAEINDIA competition to design and build a quadcopter with payload delivery, image processing, and autonomous capabilities.
- Handled systems integration, flight software development, and onboard image-processing pipelines.
- Trained a YOLOv8 model for live onboard object classification and implemented autonomous Boustrophedon path planning using ROS on hardware.

CanSat 2024 – Team Canopy

Oct 2023 – Mar 2024

- Led the conceptualization of a Can Satellite capable of ejecting a heat shield and deploying a parachute for safe payload landing from 725 m altitude.
- Managed CAD modeling, flight software programming, and ground station software development.

3D Trajectory Plotting of a Sounding Rocket

Aug 2023 – Nov 2023

- Led a 4-member team in the design and pre-flight testing of a sounding rocket equipped with a custom avionics bay for telemetry acquisition.
- Engineered a parachute ejection mechanism and contributed to flight software and PCB prototyping.
- Developed a MATLAB application to analyze IMU telemetry and generate 3D trajectory plots of the sounding rocket.

ACHIEVEMENTS

- Secured 12th rank internationally as Team Canopy in the Design Phase of CanSat 2024.
- Secured 11th rank nationally as Team Classified in AeroTHON 2024.
- Received the “Amita Award” for securing the highest score in Python Programming during sophomore year at MIT Chennai (2022).
- Received the “Giri Brothers Award” for the Best Senior Project at MIT Chennai (2025).

CERTIFICATIONS

- Autodesk Fusion 360 Simulation on Coursera (Jul 2023) – Certificate ID: 5W8JUDUJGVSF
- Drone Simulation in Simulink on Udemy (Jun 2024) – Certificate ID: UC-e80f3998-5339-407d-a5e0-be257059fa01

WORKSHOPS AND SEMINARS

- National-level seminar on “Emerging Technologies in Aerospace Developments” at MIT, Anna University – May 2024
- Flight Training Laboratory at the Indian Institute of Technology Kanpur, India – Feb 2024
- STK Software Training conducted by SS Technologies, Bangalore, India – Oct 2023
- Elements of Aeronautics with Robotics at Warsaw University of Technology, Poland – Jul 2023
- Represented the University of Colorado Boulder at the ARPA-E Energy Innovation Summit, presenting an ARPA-E funded robotics research project to industry and government stakeholders in San Diego, CA, USA – Apr 2026

PORTFOLIO

[Portfolio Link](#)